

1. Administrative & Study Identification

Posting ID: 282539189255488180 **Brief Study Title:** Study of Pembrolizumab (MK-3475) Monotherapy Versus Sacituzumab Govitecan in Combination With Pembrolizumab for Participants With Metastatic Non-small Cell Lung Cancer (NSCLC) With Programmed Cell Death Ligand 1 (PD-L1) Tumor Proportion Score (TPS) $\geq 50\%$ (MK-3475-D46) **Official Study Title:** An Open-label, Multicenter, Phase 3 Randomized, Active-Comparator-Controlled Clinical Study of Pembrolizumab (MK-3475) in Combination With Sacituzumab Govitecan Versus MK-3475 Monotherapy as First-line Treatment in Participants With PD-L1 TPS Greater than or Equal to 50% Metastatic Non-small Cell Lung Cancer (KEYNOTE-D46 / EVOKE-03) **Short Title/Acronym:** EVOKE-03 / KEYNOTE-D46 **Protocol Number:** MK-3475-D46 **NCT Number:** NCT05609968 **Trial Status:** RECRUITING (Currently recruiting participants) **Sponsor/Company Name:** Merck **Investigational Product:** Pembrolizumab (MK-3475, Trade Name: Keytruda, ATC Code: L01FF02) and Sacituzumab Govitecan (Trodelyv®) **Product Information (EMA):** [Keytruda EPAR](#) **Phase of Development:** Phase 3 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate and compare the efficacy of pembrolizumab in combination with sacituzumab govitecan versus pembrolizumab monotherapy with respect to Progression-Free Survival (PFS) and Overall Survival (OS) per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1) as assessed by blinded independent central review (BICR). **Secondary Objectives:** To evaluate Objective Response (OR) and Duration of Response (DOR), and to assess Patient-Reported Outcomes (PROs), specifically changes from baseline in the Global Health Status/Quality of Life (Items 29 and 30) Combined Score on the EORTC QLQ-C30.

2.2 Background & Rationale Non-small cell lung cancer (NSCLC) represents approximately 85% of all lung cancers, and patients presenting with metastatic disease continue to experience a relatively low 5-year relative survival rate. To address the gap in improving durable responses for treatment-naïve patients with highly expressed PD-L1 (TPS $\geq 50\%$), this study evaluates whether adding the Trop-2-directed antibody-drug conjugate (ADC) sacituzumab govitecan to the standard-of-care immunotherapy (pembrolizumab) can provide superior progression-free and overall survival compared to pembrolizumab alone.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator-controlled (Pembrolizumab alone) **Blinding:** Open-label **Randomization:** 1:1 Randomized (Parallel Assignment) **Phase Requirement:** Trial must be in PHASE3 (Phase ID: PHASE3)

3.2 Planned Enrollment & Duration Targeted Enrollment: 614 participants **Number of Sites:** Multicenter, global distribution **Estimated Study Start:** 28-Nov-2022 **Estimated Primary Completion:** [DD-MMM-YYYY] (Expected to run for up to 105 weeks per patient)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically or cytologically confirmed diagnosis of metastatic Non-Small Cell Lung Cancer (NSCLC). 3. Tumor tissue demonstrating PD-L1 Tumor Proportion Score (TPS) $\geq 50\%$ of tumor cells as assessed by immunohistochemistry (IHC) at a central laboratory. 4. Confirmation that epidermal growth factor receptor (EGFR), anaplastic lymphoma kinase 1 (ALK-1), or ROS proto-oncogene 1 (ROS-1)-directed therapy is not indicated as primary therapy. 5. Life expectancy of at least 3 months. 6. At least 1 measurable lesion per RECIST v1.1. 7. ECOG Performance Status of 0 or 1 within 7 days before randomization.

4.2 Exclusion Criteria 1. History of a second malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 3 years. 2. Previous systemic chemotherapy or other targeted or biological antineoplastic therapy for metastatic NSCLC. 3. Previously received treatment with Topoisomerase 1 inhibitors or Trop-2 targeted therapy. 4. Received prior therapy with an anti-programmed cell death 1 protein (anti-PD-1), anti-programmed cell death ligand 1 (anti-PD-L1), or anti-programmed cell death ligand 2 (anti-PD-L2) agent or with an agent directed to another stimulatory or coinhibitory T-cell receptor. 5. Received prior radiotherapy within 2 weeks of start of study intervention or has radiation-related toxicities requiring corticosteroids. 6. Received radiation therapy to the lung that is >30 Gray (Gy) within 6 months of the first dose of study intervention. 7. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. 8. Received an investigational agent or used an investigational device within 4 weeks before study intervention administration. 9. Cardiac disease, including: Myocardial infarction or unstable angina pectoris within 6 months of enrollment; History of serious ventricular arrhythmia, high-grade atrioventricular block, or other cardiac arrhythmias requiring antiarrhythmic medications; history of QT interval prolongation; New York Heart Association (NYHA) Class III or greater congestive heart failure or left ventricular ejection fraction of $<40\%$. 10. Active chronic inflammatory bowel disease. 11. Diagnosis of immunodeficiency or

is receiving chronic systemic steroid therapy or any other form of immunosuppressive therapy within 7 days prior to the first dose of study medication. 12. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 13. Severe hypersensitivity (≥ Grade 3) to pembrolizumab or sacituzumab govitecan and/or any of their excipients. 14. Active autoimmune disease that has required systemic treatment in the past 2 years except replacement therapy. 15. History of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 16. Active infection requiring systemic therapy, history of human immunodeficiency virus (HIV) infection, history of hepatitis B, or known active hepatitis C virus infection. 17. Inadequate recovery from major surgery or ongoing surgical complications. 18. History or current evidence of any condition, therapy, laboratory abnormality, or other circumstance that might confound the results of the study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm A (Experimental):** Pembrolizumab 200 mg on Day 1 + Sacituzumab govitecan 10 mg/kg on Day 1 and Day 8 of each 21-day cycle. * **Arm B (Active Comparator):** Pembrolizumab 200 mg on Day 1 of each 21-day cycle.

Route of Administration: Intravenous (IV) infusion **Duration of Treatment:** Pembrolizumab for up to 35 cycles (approximately 105 weeks). Sacituzumab govitecan continued until disease progression, unacceptable toxicity, or death.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications for symptom management (e.g., anti-emetics). **Prohibited:** Concurrent systemic antineoplastic therapy, other investigational agents (within 4 weeks), and live-attenuated vaccines (within 30 days prior).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging, Biopsy for PD-L1 Testing, Safety Labs
Baseline	Day 0	Randomization, First Dose Administration, ECOG Status Check
Interim	Every 21 Days (Q3W)	IV Infusions, Vital Signs, Safety Labs, AE Monitoring, Tumor Imaging (every 9 weeks initially)
End of Study	Up to Week 105	Final Safety Follow-up, Post-treatment Survival Follow-up (Every 9-12 weeks)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Progression-Free Survival (PFS)**: Defined as the time from randomization to the first documented progressive disease (PD) or death due to any cause, whichever occurs first as per RECIST 1.1. PD is defined as $\geq 20\%$ increase in the sum of diameters of target lesions. In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of ≥ 5 mm. The appearance of one or more new lesions is also considered PD. RECIST 1.1 is modified to follow a maximum of 10 target lesions and a maximum of 5 target lesions per organ. PFS will be assessed by Blinded Independent Central Review (BICR). 2. **Overall Survival (OS)**: Defined as the time from randomization to death due to any cause.

7.2 Secondary Endpoints 1. **Objective Response (OR)** per RECIST 1.1 by BICR. 2. **Duration of Response (DOR)** per RECIST 1.1 by BICR. 3. **Change from Baseline in the Global Health Status/Quality of Life (Items 29 and 30) Combined Score** on the European Organization for Research and Treatment of Cancer Quality of Life Questionnaire - Core 30 (EORTC QLQ-C30).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via physical exams, laboratory tests, and patient reports at every 21-day cycle visit. **Serious Adverse Events (SAEs)**: Must be reported to the sponsor and regulatory bodies within 24 hours of investigator awareness. **Data Monitoring Committee (DMC)**: An external, independent Data Monitoring Committee (eDMC) will evaluate ongoing safety and efficacy data throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary efficacy analyses; Safety Analysis Set for all subjects who receive at least one dose of the study drug. **Sample Size Justification**: Targeted Enrollment = 614 patients; powered to detect a statistically significant improvement in PFS and OS for the combination arm versus pembrolizumab monotherapy ($\alpha = 0.05$). **Handling of Missing Data**: Time-to-event variables will be censored at the last time the patient was known to be progression-free/alive (Kaplan-Meier estimates).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan**: Periodic on-site and remote monitoring

visits to ensure protocol adherence, data accuracy, and patient safety. **Audit Trail:** Mandatory utilization of validated Electronic Data Capture (EDC) systems with full audit trails and data clarification forms (DCFs).

11. Study Identifiers & Governance

Primary ID: MK-3475-D46 **Registry Number:** NCT05609968 **Sponsor/Lead Organization:** Merck **Study Title:** EVOKE-03 / KEYNOTE-D46 **Official Regulatory Title:** An Open-label, Multicenter, Phase 3 Randomized, Active-Comparator-Controlled Clinical Study of Pembrolizumab (MK-3475) in Combination With Sacituzumab Govitecan Versus MK-3475 Monotherapy as First-line Treatment in Participants With PD L1 TPS Greater than or Equal to 50% Metastatic Non-small Cell Lung Cancer

12. Clinical Framework & Ontology Mapping

Disease Areas: Carcinoma, Non-Small-Cell Lung **Primary Condition:** Carcinoma (Preferred Name: Carcinoma) **Condition Definition:** A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, those that arise from squamous epithelium are called squamous cell carcinomas, and those that arise from transitional epithelium are called transitional cell carcinomas (NCI Thesaurus). **Preferred UMLS Name:** PD-L1 Positive Non-Small-Cell Lung Carcinoma **Semantic Type:** Neoplastic Process **Diagnosis Threshold:** PD-L1 Tumor Proportion Score (TPS) $\geq 50\%$ **Medical Methodology:** Immuno-Oncology (IO) and Antibody-Drug Conjugate (ADC) Combination Therapy **Source Level:** 5

Ontology Identifiers: * **MeSH Code:** D002277 * **HPO Codes:** HP: 0030731 * **SNOMED ID:** 1187425009 * **SNOMED Hierarchy:** 1187225007 | 118285006 | 108369006 | 1240414004

13. Study Procedures & Methodology

Management Pathway: Standard first-line systemic management of metastatic PD-L1 high NSCLC **Education Protocol (HCP):** Protocol training, RECIST 1.1 assessment standardization, and AE management (specifically neutropenia/diarrhea associated with ADCs) **Education Protocol (Patient):** Treatment schedule briefing, infusion reaction reporting, and PRO questionnaire completion **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for all efficacy-related imaging

14. Observation & Follow-Up Schedule

Total Duration: Up to 105 weeks of active treatment followed by indefinite survival follow-up **Onsite Visit Cadence:** Every 3 weeks (Q3W) matching the 21-day treatment cycles **Remote Monitoring Frequency:** Routine data monitoring intervals per sponsor clinical operations plan **Checklist Review Interval:** Imaging assessments generally every 9 weeks

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					